The documentation and process conversion measures necessary to comply with this revision shall be completed by 2 January 1995

INCH-POUND

MIL-S-19500/560B
2 September 1994
SUPERSEDING
MIL-S-19500/560A
31 March 1993

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, SWITCHING
TYPE 2N5339 JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by US Army Laboratory Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN, silicon, switching transistors. Three levels of product assurance are provided for each device type as specified in MIL-S-19500. Two levels of product assurance are provided for each unencapsulated device type. See 6.5 for JAN level.

1.2 Physical dimensions. See figure 1 (TO-39) and figure 2 for JANHC and JANKC (die) dimensions.

1.3 Maximum ratings. Unless otherwise specified, $T_A$ = +25°C.

| $P_T$ 1/ $T_A$ = 25°C | $P_T$ 2/ $T_C$ = 25°C | $V_{CBO}$ | $V_{CEO}$ | $V_{EBO}$ | $I_C$ | $I_B$ | $T_{STG}$ and $T_J$ |
|---|---|---|---|---|---|---|---|
| W | W | V dc | V dc | V dc | A dc | A dc | °C |
| 1.0 | 10 | 100 | 100 | 6.0 | 5.0 | 1.0 | -65 to +200 |

1/ Derate linearly 5.71 mW/°C for $T_A$ > 25°C.
2/ Derate linearly 57.1 mW/°C for $T_C$ > 25°C.

1.4 Primary electrical characteristics at $T_A$ = 25°C.

| Limits | $h_{FE}$ at $V_{CE}$ = 2.0 V dc 1/ | | | |
|---|---|---|---|---|
| | $(h_{FE1})$ $I_C$ = 0.5 A dc | $(h_{FE2})$ $I_C$ = 2.0 A dc | $(h_{FE3})$ $I_C$ = 5.0 A dc | $R_{\theta JC}$ |
| | | | | °C/W |
| Min | 60 | 60 | 40 | |
| Max | | 240 | | 17.5 |

1/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Laboratory Command, Electronics Technology and Devices Laboratory, ATTN: SLCET-RS, Fort Monmouth, NJ 07703-5601, by using the Standardization Document Improvement proposal (DD Form 1426 appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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| Limits | $\lvert h_{fe} \rvert$<br>f = 10 MHz<br>$V_{CE}$ = 10 V dc<br>$I_C$ = 0.5 A dc | $C_{obo}$<br>100 kHz ≤ f < MHz<br>$V_{CB}$ = 10 V dc<br>$I_E$ = 0 | Switching<br>$t_{on}$<br>See fig. 4 | Switching<br>$t_{off}$<br>See fig. 5 | $V_{CE(sat)1}$ 1/<br>$I_C$ = 2.0 A dc<br>$I_B$ = 0.2 A dc | $V_{BE(sat)1}$ 1/<br>$I_C$ = 2.0 A dc<br>$I_B$ = 0.2 A dc |
|---|---|---|---|---|---|---|
| | | pF | µs | µs | V dc | V dc |
| Min | 3.0 | | | | | |
| Max | 15 | 250 | 0.2 | 2.2 | 0.7 | 1.2 |

1/ Pulsed (see 4.5.1).

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

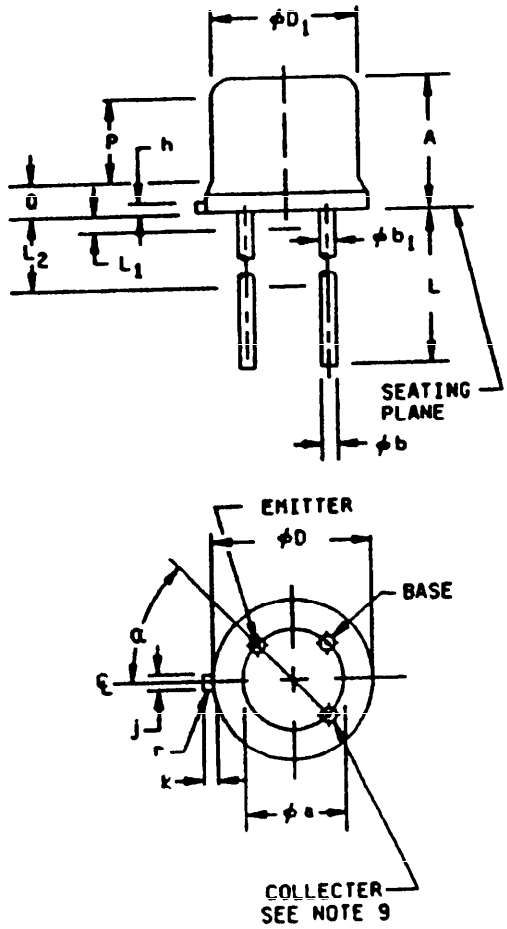
3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and on figures 1 and 2 herein.

3.3.1 Lead material and finish. Lead material shall be Kovar or Alloy 52. Lead finish shall be gold-plated, tin-plated, or solder dipped. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).



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| Dimensions | | | | | |
|---|---|---|---|---|---|
| Ltr | Inches | | Millimeters | | Notes |
| | Min | Max | Min | Max | |
| A | .240 | .260 | 6.10 | 6.60 | |
| $\phi$a | .200 TP | | 5.08 TP | | 6 |
| $\phi$b | .016 | .021 | 0.41 | 0.53 | 7,6 |
| $\phi b_1$ | .016 | .019 | 0.41 | 0.48 | 7,8 |
| $\phi$D | .335 | .370 | 8.51 | 9.40 | |
| $\phi D_1$ | .305 | .335 | 7.75 | 8.51 | |
| h | .009 | .041 | 0.23 | 1.04 | |
| j | .028 | .034 | 0.71 | 0.86 | 2 |
| k | .029 | .045 | 0.74 | 1.14 | 3 |
| L | .500 | .750 | 12.70 | 19.05 | 7,8 |
| $L_1$ | | .050 | | 1.27 | 7,8 |
| $L_2$ | .250 | - | 6.35 | | 7,8 |
| P | .100 | | 2.54 | | 5 |
| Q | | .050 | | 1.27 | 4 |
| r | | .010 | | 0.25 | 10 |
| $\alpha$ | 45°TP | | 45° TP | | 6 |

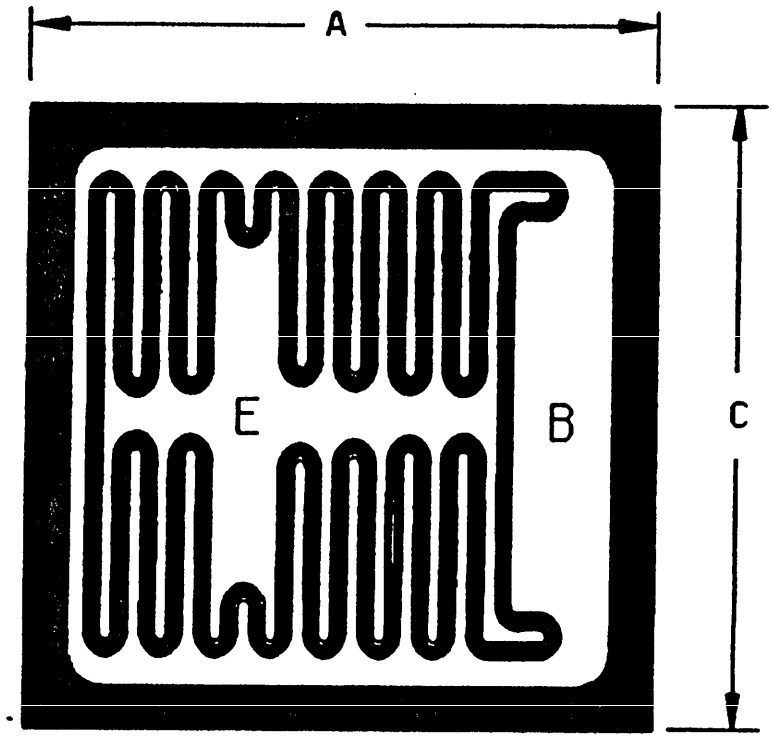



NOTES:
1. Metric equivalents are given for general information only.
2. Beyond radius (r) maximum, j shall be held for a minimum length of .011 inch (0.28 mm).
3. Dimension k measured from maximum $\phi$D.
4. Outline in this zone is not controlled.
5. Dimension $\phi D_1$ shall not vary more than .010 inch (0.25 mm) in zone P. This zone is controlled for automatic handling.
6. Leads at gauge plane .054 +.001, -.000 inch (1.37 +0.03, -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to table at MMC. The device may be measured by direct methods or by the gauge and gauging procedure shown on figure 3.
7. $\phi b_1$ applies between $L_1$ and $L_2$. $\phi$b applies between $L_2$ and L minimum. Diameter is uncontrolled in $L_1$ and beyond L minimum.
8. All three leads.
9. The collector shall be electrically and mechanically connected to the case.
10. Radius (r) applies to both inside corners of tab.
11. Dimensions are in inches.

FIGURE 1. Physical dimensions (TO-39).



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A version

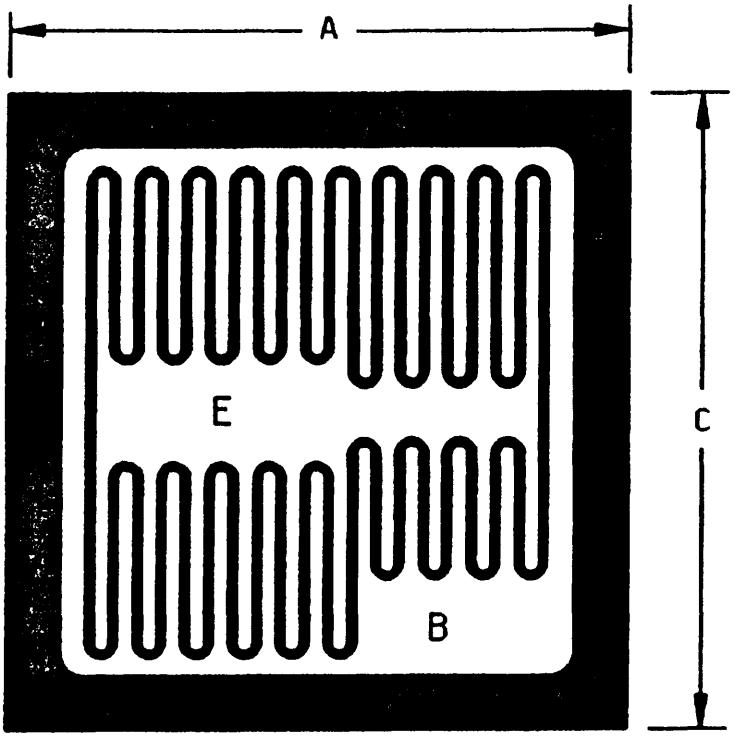
| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| A | .120 | .124 | 3.05 | 3.15 |
| C | .120 | .124 | 3.05 | 3.15 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The physical characteristics of the die are:
   Thickness: .008 inch (0.20 mm) to .012 inch (0.30 mm).
   Top metal: Aluminum 40,000 Å minimum, 50,000 Å nominal.
   Back metal: Gold 2,500 Å minimum, 3,000 Å nominal.
   Back side: Collector.
   Bonding pad: B = .015 inch (0.38 mm) x .072 inch (1.83 mm).
   E = .015 inch (0.38 mm) x .060 inch (1.52 mm).

FIGURE 2. Physical dimensions JANHCA and JANKCA die.



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B version

| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| A | .10 | .10 | 2.54 | 2.54 |
| C | .10 | .10 | 2.54 | 2.54 |

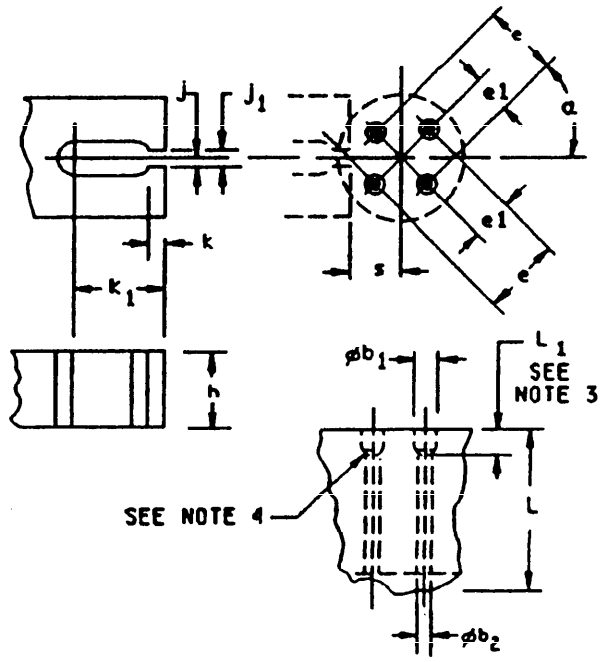
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The physical characteristics of the die are:
   Thickness: .006 inch (0.15 mm) to .010 inch (0.25 mm).
   Top metal: Aluminum 25,000 Å minimum, 33,000 Å nominal.
   Back metal: Gold 1,500 Å minimum, 2,000 Å nominal.
   Back side: Collector.

FIGURE 2. Physical dimensions JANHCB and JANKCB die - Continued.



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| Dimensions | | | | | |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| Ltr | Min | Max | Min | Max | Notes |
| $\phi b_1$ | .0595 | .0605 | 1.511 | 1.537 | |
| $\phi b_2$ | .0325 | .0335 | 0.826 | 0.851 | |
| e | .1995 | .2005 | 5.067 | 5.093 | |
| $e_1$ | .0995 | .1005 | 2.527 | 2.553 | |
| h | .150 Nominal | | 3.81 Nominal | | |
| j | .0175 | .0180 | 0.445 | 0.457 | |
| $J_1$ | .0350 | .0355 | 0.889 | 0.902 | |
| k | .009 | .011 | 0.23 | 0.28 | |
| $k_1$ | .125 Nominal | | 3.18 Nominal | | |
| L | .372 | .378 | 9.45 | 9.60 | |
| $L_1$ | .054 | .055 | 1.37 | 1.40 | |
| s | .182 | .199 | 4.62 | 5.05 | 3 |
| α | 44.90° | 45.10° | 44.90° | 45.10° | |




NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The location of the tab locator within the limits indicated will be determined by the tab and flange dimensions of the device being checked.
4. Gauging procedure. The device being measured shall be inserted until its seating plane is .125 ±.010 inch (3.18 ±0.25 mm) from the seating surface of the gauge. A force of 8 ±.5 ounce shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gauge. The use of a pin straightener prior to insertion in the gauge is permissible.
5. Gauging plane.
6. Drill angle.

FIGURE 3. Gauge for lead and tab locations.



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3.4 Marking. Marking shall be in accordance with MIL-S-19500. At the option of the manufacturer, the following marking may be omitted from the body of the transistor:

a. Country of origin.

b. Manufacturer's identification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500 and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.2.1 Group E. Group E inspection shall be conducted in accordance with MIL-S-19500 and table II herein.

4.3 Screening JANS, JANTX, and JANTXV levels only. Screening shall be in accordance with table II of MIL-S-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

4.3.1 Screening (JANHC and JANKC). Screening of die shall be in accordance with MIL-S-19500 appendix H.

4.3.2 JANHC and JANKC die. Qualification shall be in accordance with MIL-S-19500.

| Screen (see table II of MIL-S-19500) | Measurement | |
|---|---|---|
| | JANS level | JANTX and JANTXV levels |
| 9 | $I_{CBO}$ and $h_{FE2}$ | Not applicable |
| 11 | $I_{CBO1}$; $h_{FE2}$; $\Delta I_{CBO}$ = 100 percent of initial value or 1.0 μA dc, whichever is greater; $\Delta h_{FE2}$ = ±15 percent of initial value. | $I_{CBO}$ and $h_{FE2}$ |
| 12 | See 4.3.2 | See 4.3.2 |
| 13 | Subgroups 2 and 3 of table I herein; $\Delta I_{CBO}$ = 100 percent of initial value or 1.0 μA dc, whichever is greater; $\Delta h_{FE2}$ = ±15 percent initial value. | Subgroup 2 of table I herein; $\Delta I_{CBO}$ = 100 percent of initial value or 1.0 μA dc, whichever is greater; $\Delta h_{FE}$ = ±15 percent initial value. |

4.3.2 Power burn-in conditions. Power burn-in conditions are as follows:

20 V dc ≥ $V_{CB}$ ≥ 10 V dc, $P_T$ = 1.0 W at $T_A$ = 25°C ±5°C.

NOTE: No heat sink or forced air cooling on the devices shall be permitted.



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4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein. Electrical measurements (end-points) shall be in accordance with the inspections of table I, group A, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and IVb (JANTX and JANTXV) of MIL-S-19500 and as follows. Electrical measurements (end-points) shall be in accordance with the inspections of table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

| Subgroup | Method | Condition |
|---|---|---|
| 4 | 1037 | $20 \geq V_{CB} \geq 10$ V dc, $P_T = 1.0$ W at $T_A = +25°C \pm 3°C$; $t_{on} = t_{off} = 3$ minutes minimum for 2,000 cycles. No heat sink or forced air on the device shall be permitted. |
| 5 | 1027 | $V_{CB} = 10$ V dc, $P_T = 1.6$ W at $T_A = +25°C$ for 96 hours. No heat sink or forced air cooling on the devices shall be permitted. |
| 5 | 2037 | Test condition A. |

4.4.2.2 Group B inspection, (table IVb of MIL-S-19500).

| Subgroup | Method | Condition |
|---|---|---|
| 3 | 1027 | $V_{CB} \geq 10$ V dc, $P_T = 1.0$ W, $T_A = +25°C \pm 3°C$. No heat sink or forced air cooling on devices shall be permitted. |
| 6 | 1032 | $T_A = +200°C$. |
| 7 | | High-temperature operation; $T_A = +150°C$. Collector to emitter voltage (saturated); method 3071; $I_C = 2.0$ A dc, $I_B = 0.2$ A dc, (pulsed, see (4.5.1); $\Delta V_{CE(sat)3}$, 0.75 V dc maximum. |
| | | Low-temperature operation; $T_A = -55°C$. Base to emitter voltage (saturated); method 3066; test condition A, $I_C = 2.0$ A dc, $I_B = 0.2$ A dc; pulsed (see 4.5.1); $\Delta V_{BE(sat)3}$ 1.3 V dc maximum. |

4.4.3. Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500 and as follows. Electrical measurements (end-points) shall be in accordance with the inspections of table I, group A, subgroup 2 herein.

4.4.3.1 Group C inspection, (table V of MIL-S-19500).

| Subgroup | Method | Condition |
|---|---|---|
| 2 | 2036 | Test condition E. |
| 6 | 1026 | $V_{CB} \geq 10$ V dc, $P_T = 1.0$ W, $T_A = +25°C \pm 3°C$. No heat sink or forced cooling on the devices shall be permitted. |



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical inspection | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Breakdown voltage, collector to emitter | 3011 | Bias condition D; $I_C$ = 50 mA dc; pulsed (see 4.5.1) | $V_{(BR)CEO}$ | 100 | | V dc |
| Collector to emitter cutoff current | 3041 | Bias condition D; $V_{CE}$ = 100 V | $I_{CEO}$ | | 100 | µA dc |
| Collector to emitter cutoff current | 3041 | Bias condition A; $V_{CE}$ = 90 V dc; $V_{BE}$ = +1.5 V dc | $I_{CEX1}$ | | 10 | µA dc |
| Collector to base cutoff current | 3036 | Bias condition D; $V_{CB}$ = 100 V dc | $I_{CBO}$ | | 10 | µA dc |
| Emitter to base cutoff current | 3061 | Bias condition D; $V_{EB}$ = 6.0 V dc | $I_{EBO}$ | | 100 | µA dc |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 2.0 V dc; $I_C$ = 0.5 A dc; pulsed (see 4.5.1) | $h_{FE1}$ | 60 | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 2.0 V dc; $I_C$ = 2.0 A dc; pulsed (see 4.5.1) | $h_{FE2}$ | 60 | 240 | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 2.0 V dc; $I_C$ = 5.0 A dc; pulsed (see 4.5.1) | $h_{FE3}$ | 40 | | |
| Collector to emitter voltage (saturated) | 3071 | $I_C$ = 2.0 A dc; $I_B$ = 0.2 A dc; pulsed (see 4.5.1) | $V_{CE(sat)1}$ | | 0.7 | V dc |
| Collector to emitter voltage (saturated) | 3071 | $I_C$ = 5.0 A dc; $I_B$ = 0.5 A dc; pulsed (see 4.5.1) | $V_{CE(sat)2}$ | | 1.2 | V dc |
| Base emitter voltage (saturated) | 3066 | Test condition A; $I_C$ = 2.0 A dc; $I_B$ = 0.2 A dc; pulsed (see 4.5.1) | $V_{BE(sat)1}$ | | 1.2 | V dc |
| Base emitter voltage (saturated) | 3066 | Test condition A; $I_C$ = 5.0 A dc; $I_B$ = 0.5 A dc; pulsed (see 4.5.1) | $V_{BE(sat)2}$ | | 1.8 | V dc |

See footnote at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 3 | | | | | | |
| High-temperature operation: | | $T_A$ = +150°C | | | | |
| Collector to base cutoff current | 3041 | Bias condition A; $V_{CE}$ = 90 V dc; $V_{BE}$ = +1.5 V dc | $I_{CEX2}$ | | 1.0 | mA dc |
| Low-temperature operation: | | $T_A$ = -55°C | | | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 2.0 V dc; $I_C$ = 2.0 A dc; | $h_{FE4}$ | 12 | | |
| Subgroup 4 | | | | | | |
| Magnitude of small-signal short-circuit forward-current transfer ratio | 3306 | $V_{CE}$ = 10 V dc; $I_C$ = 0.5 A dc; f = 10 MHz | $\|h_{fe}\|$ | 3 | 15 | |
| Open circuit output capacitance | 3236 | $V_{CB}$ = 10 V dc; $I_E$ = 0; 100 kHz ≤ f ≤ 1 MHz | $C_{obo}$ | | 250 | pF |
| Input capacitance (output open-circuited) | 3240 | $V_{BE}$ = 2.0 V dc; $I_C$ = 0; 100 kHz ≤ f ≤ 1 MHz (see 4.5.2) | $C_{ibo}$ | | 1000 | pF |
| Pulse response | | | | | | |
| Pulse delay time | | See figure 4 | $t_d$ | | 100 | ns |
| Pulse rise time | | See figure 4 | $t_r$ | | 100 | ns |
| Pulse storage time | | See figure 5 | $t_s$ | | 2 | µs |
| Pulse fall time | | See figure 5 | $t_f$ | | 200 | ns |
| Subgroup 5 | | | | | | |
| Safe operating area (continuous dc) | 3051 | $T_C$ = 25°C; t ≥ 0.5 s; 1 cycle | | | | |
| Test 1 | | $I_C$ = 5.0 A dc; $V_{CE}$ = 2 V dc | | | | |
| Test 2 | | $I_C$ = 2.0 A dc; $V_{CE}$ = 5 V dc | | | | |
| Test 3 | | $I_C$ = 55 mA dc; $V_{CE}$ = 90 V dc | | | | |
| Electrical measurements | | See table I, group A, Subgroup 2 | | | | |
| Subgroups 6 and 7 | | | | | | |
| Not applicable | | | | | | |

1/ For sampling plan, see MIL-S-19500.



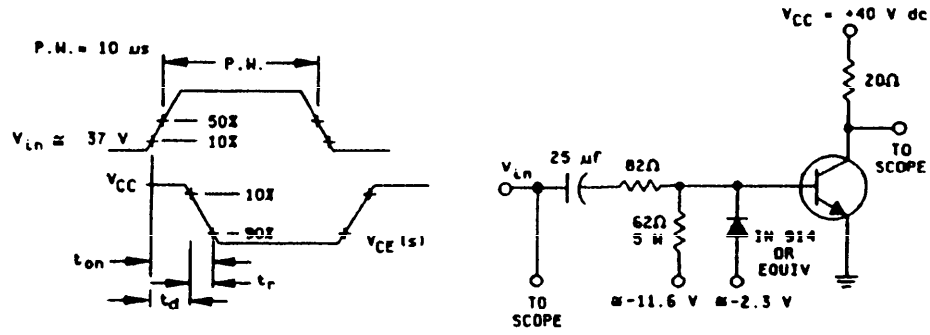
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TABLE II. Group E inspection (all quality levels) for qualification only.

| Inspection | Method | MIL-STD-750 Conditions | Qualification and large lot quality conformance inspection |
|---|---|---|---|
| Subgroup 1 | | | 22 devices, c = 0 |
| Temperature cycling | 1051 | 500 cycles | |
| Electrical measurements | | See table I, group A, subgroup 2 | |
| Subgroup 2 | | | 22 devices, c = 0 |
| Steady-state dc | 1039 | Condition A, 340 hours | |
| Electrical measurements | | See table I, group A, subgroup 2 | |
| Subgroup 3 | | | |
| Not applicable | | | |
| Subgroup 4 | | | 22 devices, c = 0 |
| Thermal resistance | 3131 | See 4.5.3 | |
| Subgroup 5 | | | |
| Not applicable | | | |

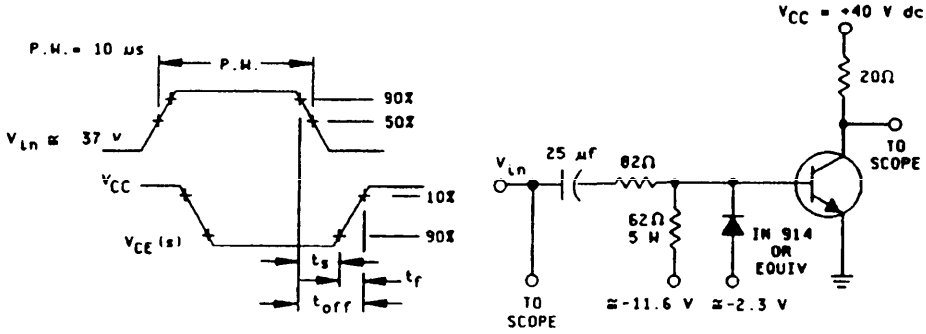


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NOTES:
1. The rise time ($t_r$) of the applied pulse shall be ≤ 20 ns, duty cycle ≤ 2 percent, and the generator source impedance shall be 50Ω.
2. Scope input: $Z_{IN}$ ≥ 1 MΩ, $C_{IN}$ ≤ 20 pF, rise time ≤ 20 ns.
3. $t_{on}$ conditions: $I_C$ = 2A, $I_{B1}$ = 200 mA.

FIGURE 4. Saturated turn-on switching waveform and time test circuit.




NOTES:
1. The rise time ($t_r$) of the applied pulse shall be ≤ 20 ns, duty cycle ≤ 2 percent, and the generator source impedance shall be 50Ω.
2. Scope input: $Z_{IN}$ ≥ 1 MΩ, $C_{IN}$ ≤ 20 pF, rise time ≤ 20 ns.
3. $t_{off}$ conditions: $I_C$ = 2A, $I_{B1}$ = $-I_{B2}$ = 200 mA.

FIGURE 5. Saturated turn-off switching time waveform and test circuit.



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4.5 Methods of inspection. Methods of inspection shall be as specified in appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Input capacitance. This test shall be conducted in accordance with method 3240 of MIL-STD-750, except the output capacitor shall be omitted.

4.5.3 Thermal resistance (to be performed for qualification inspection only). Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power applications shall be .15 A dc.

b. Collector to emitter voltage magnitude shall be 20 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference point temperature shall be 25°C $\leq T_R \leq$ 35°C and recorded before the test is started.

e. Mounting arrangement shall be with heat sink to case.

f. Maximum limit shall be $R_{\theta JC}$ = 17.5°C/W.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.1).

c. For die acquisition, specify the JANHC or JANKC letter version (see figures 3, 4, and 5).

d. Product assurance level and type designation.

6.3 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's Part or Identifying Number (PIN). This information in no way implies that manufacturer's PIN's are suitable as a substitute for the military PIN.

6.4 Application guidance. The following PNP type transistor is complimentary to the NPN device listed herein.

| NPN | PNP |
|---|---|
| 2N5339 | 2N6193 |



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6.5 Replacement data. JANTX devices shall be a direct replacement for JAN devices (example: JANTX2N5339 and JAN2N5339. Manufacturers can no longer manufacture device types from this specification with a JAN assurance level.

6.6 Suppliers of JANHC and JANKC die. The qualified die suppliers with the applicable letter version (example JANHCA2N5339) will be identified on the QPL.

| JANC ordering information | | |
|---|---|---|
| PIN | Manufactures | |
| | 33178 | 34156 |
| 2N5339 | JANHCA2N5339<br>JANKCA2N5339 | JANHCB2N5339<br>JANKCB2N5339 |

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodian:
Army - ER

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5961-1684)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

**INSTRUCTIONS**

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER<br>MIL-S-19500/560B | 2. DOCUMENT DATE (YYMMDD)<br>940902 |
|---|---|---|

**3. DOCUMENT TITLE**
SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, SWITCHING TYPE 2N5339 JANTX, JANTXV, JANS, JANHC, AND JANKC

**4. NATURE OF CHANGE** (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

**5. REASON FOR RECOMMENDATION**

**6. SUBMITTER**

| a. NAME (Last, First, Middle initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON<br>(If applicable) | 7. DATE SUBMITTED<br>(YYMMDD) |

**8. PREPARING ACTIVITY**

| a. NAME<br>DIRECTOR, US ARMY RESEARCH LAB/EPSD | b. TELEPHONE (Include Area Code)<br>(1) Commercial: 908-544-3441/3148<br>(2) AUTOVON: 995-3441/3148 |
|---|---|
| c. ADDRESS (Include Zip Code)<br>ATTN: AMSRL-EP-RC<br>FORT MONMOUTH, NJ 07703-5601 | **IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:**<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 |



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